

Maintaining healthy soils is important for agricultural production, essential ecosystem services and resilience to climate change. Healthy soils help to:

- promote farm productivity by supporting nutrients for plant growth and higher yields
- enhance water and nutrient use efficiency by reducing the risk of erosion and surface run off
- reduce emissions through carbon storage.

What you can do

Soil management practices can improve soil health and productivity. You can explore:

- checking your soil health
- managing your soil.

Check your soil health

You may want to start by testing your soil to understand its physical, chemical and biological issues. This information can be used to plan for managing:

- physical issues like compaction, poor structure, or drainage
- chemical problems with pH, salinity, or nutrient imbalances
- biological issues like lack of helpful microorganisms.

Testing your soil can also help:

- manage soil structure to help water infiltration and retention, while also keeping the soil aerated
- plan efficient fertiliser use so it matches the needs of your crops or pasture
- track soil organic matter and carbon over time.

Find more information on [soil testing](#) (Agriculture Victoria) and [explore resources](#) (Soil Science Australia).

Manage your soil

Many Australian farms are already using a mix of approaches to manage soil. It's likely that you're already familiar with the options, and you might even be using them on your land.

Choosing the best approach for you will depend on:

- your soil type
- the local climate
- what you farm
- your goals for managing your farm.

Many methods for [improving your soil health can also increase carbon storage](#) (CSIRO) to help manage your farm's emissions.

Add soil amendments

Soil testing will help you decide what nutrients are needed and in what form. Soil amendments are materials added to the soil to manage soil issues and improve soil health.

To improve your soil, you may want to:

- Use organic materials like manure, crop residues, biosolids, or compost when practical and cost-effective. These add nutrients, carbon, and beneficial microbes to the soil.
- Apply other nutrients like lime to correct soil acidity and gypsum to help improve compaction, surface crusting or high sodium.

It is important to check the quality of any products you use and make sure claimed benefits are backed by solid research.

Learn more about [soil amendments](#) (Soil Quality Knowledge Base).

Efficient fertiliser use can also improve your soil, see more information on improving fertiliser use in [improve productivity](#).

Adopt management practices

There are a range of practices that you can explore to improve soil health. These include:

- Using no-till, minimum till or strategic tillage systems to reduce soil disturbance. This can also help avoid soil erosion, conserve soil moisture and strengthen overall soil resilience.
- Maintaining stubble to increase moisture and nutrient content of soil and reduce the risk of soil erosion. Stubble retention can also reduce your farm's emissions by avoiding the need to burn residues and returning carbon from plant matter back to the soil as it rots.
- Maintaining continuous plant cover where possible, such as green or brown manure crops between seasons. This helps protect the soil, adds organic matter, uses available nitrogen, and prevents nutrient loss through leaching.
- Rotating crops and include perennial pastures and legumes in rotations. Perennials generally improve or stabilise soil carbon better than annuals.
- Avoiding overgrazing pastures and keep sufficient groundcover. Consider stock containment areas to protect pastures and groundcover.
- Managing livestock movement and paddock rotations to spread manure evenly and reduce soil compaction from hooves.

This is not an exhaustive list. Learn more about these practices and other ways to improve your soil:

- [NSW Government](#): Ways to improve your soil and build soil carbon
- [Meat & Livestock Australia](#): Managing soil to promote plant and animal health and sustain optimal biological productivity
- [Soils CRC](#): Ways to improve soil health
- [Soils for Life](#): Cropping practice guides for soil health

Some of the above management practices may also be eligible activities under the Australian Carbon Credit Unit Scheme (ACCU).

Register soil carbon projects with the ACCU Scheme

If you are looking to increase carbon stored in soil, you might be eligible to register a project with the [Australian Carbon Credit Unit \(ACCU\) Scheme](#).

Before participating in the scheme, it's important to think about whether it suits your goals and situation. Every farm and business are different, so what works for one may not work for another.

Find information on [understanding soil carbon storage potential](#) (Soil CRC) and what activities qualify under the [ACCU Scheme](#) (Clean Energy Regulator).

Resources to help you improve soil health

Explore further key resources on improving your soil health.

State and territory resources

State and territory governments have resources with information and guides on how to improve your soil health:

- New South Wales: [Soil management guides](#)
- Northern Territory: [Resource management for growers](#)
- Queensland: [Soil management](#)
- South Australia: [Soil and land management](#)
- Tasmania: [Soil management](#)
- Victoria: [Understanding soil and soil management](#)
- Western Australia: [Managing soils](#)

See more:

Websites that can help you to learn more about soil health and resilience:

- [Agriculture Victoria](#): Learn more about soil carbon and management practises.
- [CSIRO](#): Learn about research on how to improve soil management.
- [CSIRO](#): Learn more about the National Soil Monitoring Program.
- [DAFF](#): Learn more about the relationship between agricultural land management practises and soil condition.
- [DAFF](#): Learn about the national approach to protecting and managing soil.
- [Soil CRC](#): Learn more about how to improve the performance of your soils.